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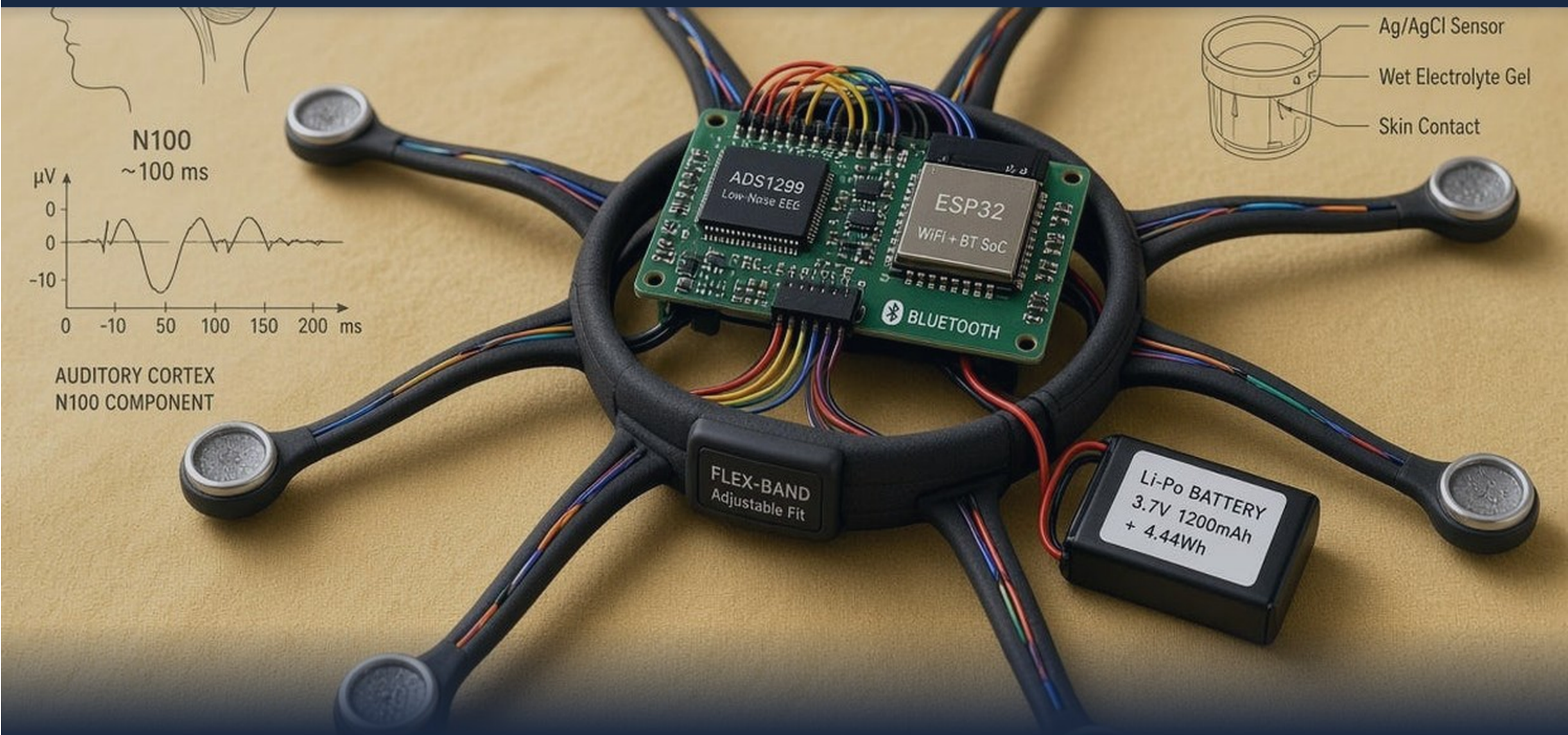
PROJECT PLAN

Building the headset, funding the work, running the study



A three-year plan for the programme the v4.2 experiment design describes: fifty-nine tasks across seven workstreams, from a fabrication pack that is finished today to a replication cohort in 2029.

> **What this plan commits to**
>
> Dates, owners and rates are stated. Every number that belongs to a supplier is marked PLACEHOLDER and stays that way until the RFQ responses land.

Stephane van der Aa
Brussels · September 2026

Companion to the experiment design v4.2

PROJECT PLAN

TI One Voice — EEG Study

September 2026 to August 2029

Version 1.0 · 2 September 2026 · Brussels, Belgium ·

How to read this plan

Seven workstreams, fifty-nine tasks, three years. Work already finished is marked complete and dated. Every cost that belongs to a supplier is written PLACEHOLDER — RFQ-EEG-001 Rev E is the document that fills those in, and guessing them here would only make the total look more settled than it is. The companion document is the experiment design v4.2, which this plan schedules rather than restates.

Project statement

Some people report hearing voices they experience as transmitted from outside their own head. Whatever the cause, the reports are consistent, specific and largely untested. This programme asks a narrow, answerable question about them: can a low-cost EEG headset distinguish, in the recorded signal, between speech a person produces, speech they hear from a loudspeaker, and the voices they report hearing when no external source is present?

The question is deliberately modest. It does not ask what causes the voices, and it does not assume an answer. It asks whether the three conditions look different to a classifier that was trained on the first two and has never seen the third. A clear negative is as publishable as a clear positive, and both are worth more than the silence the question currently receives.

Delivering it needs three things that do not yet exist together: an instrument cheap enough to put in many homes and honest enough to trust, a software platform that can take a session from a participant's laptop to a defensible dataset, and an academic partnership that gives the work an ethics committee, a statistician and a route to publication. This plan covers all three, over three years, in three study stages.

The instrument design is substantially complete and is the subject of an open RFQ. The platform is specified and its production release is targeted for January 2027. The partnership is the outstanding item and is the highest-value thing the programme can secure, because it converts several of the largest cost lines in this plan from purchases into contributions.

What this plan does not do is pretend to certainty it has not earned. Every manufacturing cost is a placeholder until the RFQ responses arrive. Every academic and clinical rate is stated at both what it costs if contributed and what it costs if bought. Nothing has been manufactured, nothing has been measured on hardware, and the schedule reflects that rather than assuming it away.

Programme schedule

Seven workstreams over three years. The near-term critical path is funding, the first prototype build, and the platform's initial production release in January 2027 — everything in the study stages waits on those three. The chart runs over two pages on a shared axis; the red line is today.

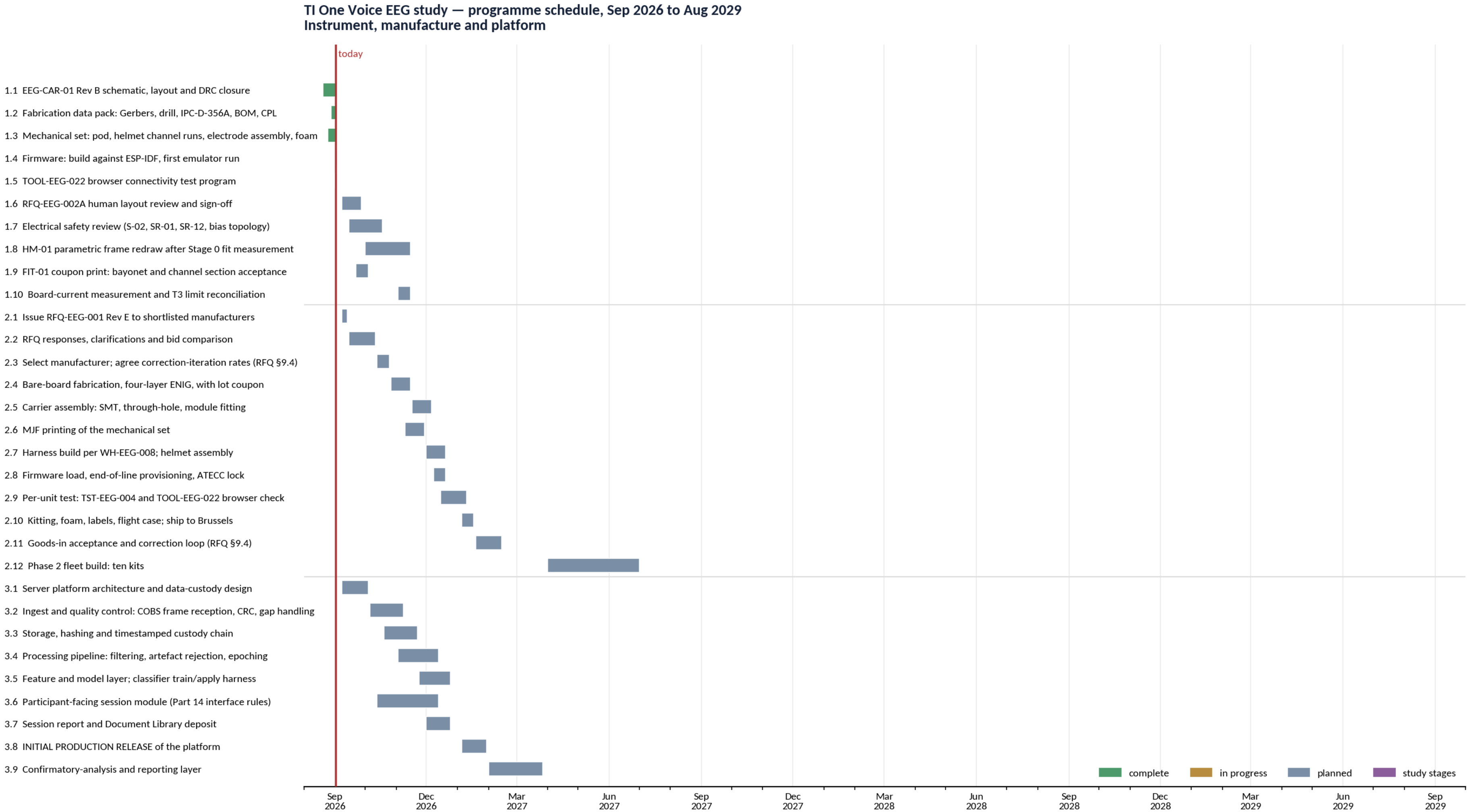


Figure 1 — Instrument, manufacture and platform. Study stages carry their own colour because they are gated on everything to their left.

Programme schedule, continued

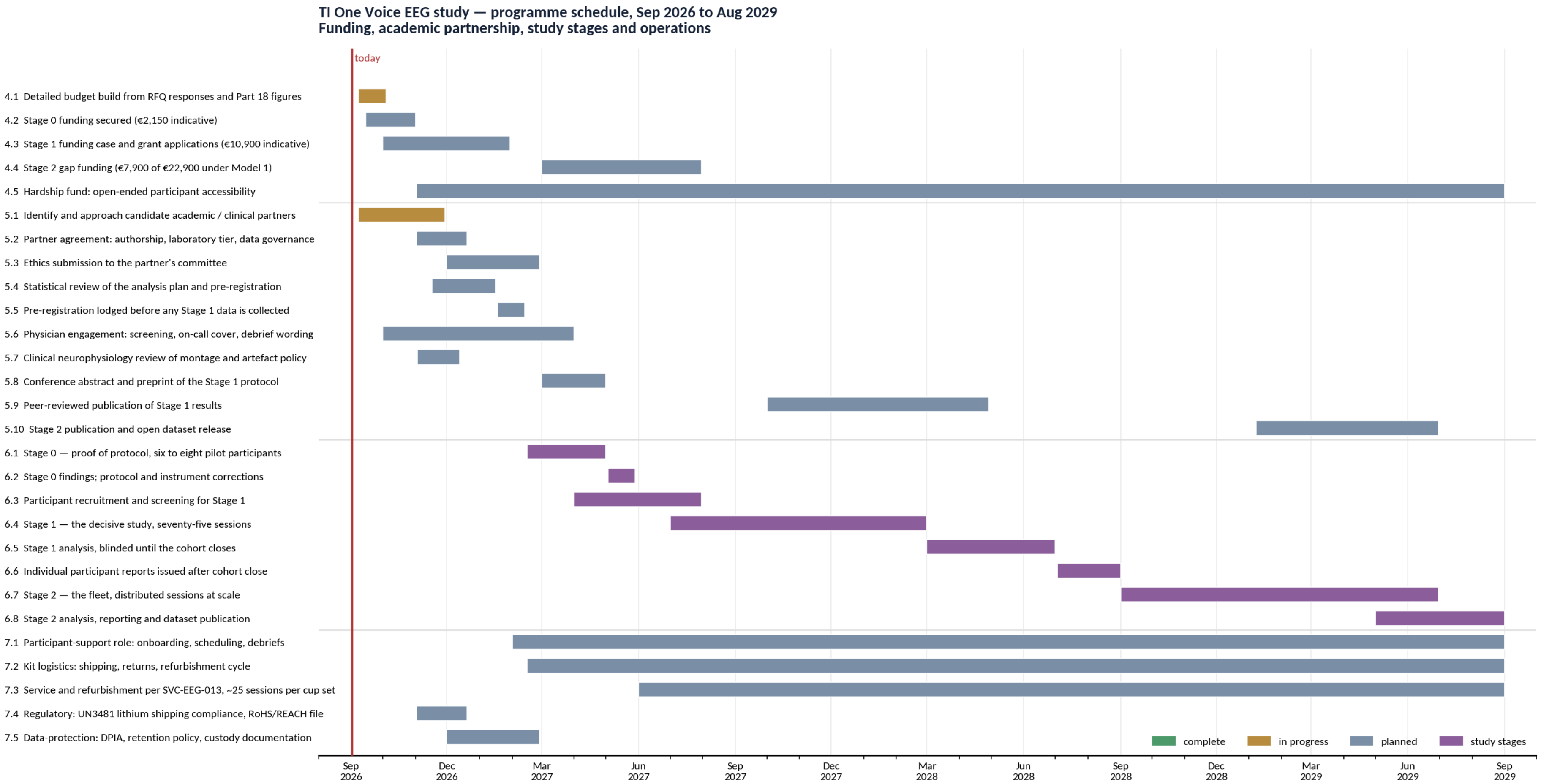


Figure 2 — Funding, academic partnership, study stages and operations. Study stages carry their own colour because they are gated on everything to their left.

Task list

Every deliverable produced so far, and everything the study needs for the next three years. Dates are planning dates. Cost entries marked PLACEHOLDER are deliberately unpriced: they belong to the RFQ-EEG-001 responses and will be filled in when those arrive.

Instrument

The EEG kit itself: carrier board, firmware, mechanical set and the browser test program. Most of this is complete; what remains needs a reviewer, a printer or a bench.

ID	Task	Start	End	Status	Owner	Cost basis
1.1	EEG-CAR-01 Rev B schematic, layout and DRC closure	20 Aug 26	02 Sep 26	complete	Engineering (in-kind)	—
1.2	Fabrication data pack: Gerbers, drill, IPC-D-356A, BOM, CPL	28 Aug 26	02 Sep 26	complete	Engineering (in-kind)	—
1.3	Mechanical set: pod, helmet channel runs, electrode assembly, foam	25 Aug 26	02 Sep 26	complete	Engineering (in-kind)	—
1.4	Firmware: build against ESP-IDF, first emulator run	02 Sep 26	02 Sep 26	complete	Engineering (in-kind)	—
1.5	TOOL-EEG-022 browser connectivity test program	01 Sep 26	02 Sep 26	complete	Engineering (in-kind)	—
1.6	RFQ-EEG-002A human layout review and sign-off	08 Sep 26	26 Sep 26	planned	External layout engineer	PLACEHOLDER — RFQ response
1.7	Electrical safety review (S-02, SR-01, SR-12, bias topology)	15 Sep 26	17 Oct 26	planned	Electrical-safety reviewer	€60–90/h loaded, €90–150/h consultancy
1.8	HM-01 parametric frame redraw after Stage 0 fit measurement	01 Oct 26	14 Nov 26	planned	Mechanical engineer (volunteer)	In-kind, else €60–90/h
1.9	FIT-01 coupon print: bayonet and channel section acceptance	22 Sep 26	03 Oct 26	planned	MJF print bureau	PLACEHOLDER — print quote
1.10	Board-current measurement and T3 limit reconciliation	03 Nov 26	14 Nov 26	planned	Engineering (in-kind)	—

Manufacture

From issuing the RFQ to accepting tested kits in Brussels. Every cost here is a placeholder until the responses arrive.

ID	Task	Start	End	Status	Owner	Cost basis
2.1	Issue RFQ-EEG-001 Rev E to shortlisted manufacturers	08 Sep 26	12 Sep 26	planned	Programme lead	—
2.2	RFQ responses, clarifications and bid comparison	15 Sep 26	10 Oct 26	planned	Programme lead	—
2.3	Select manufacturer; agree correction-iteration rates (RFQ §9.4)	13 Oct 26	24 Oct 26	planned	Programme lead	PLACEHOLDER — RFQ response
2.4	Bare-board fabrication, four-layer ENIG, with lot coupon	27 Oct 26	14 Nov 26	planned	PCB fabricator	PLACEHOLDER — RFQ line
2.5	Carrier assembly: SMT, through-hole, module fitting	17 Nov 26	05 Dec 26	planned	EMS / assembly house	PLACEHOLDER — RFQ line
2.6	MJF printing of the mechanical set	10 Nov 26	28 Nov 26	planned	MJF print bureau	PLACEHOLDER — RFQ line
2.7	Harness build per WH-EEG-008; helmet assembly	01 Dec 26	19 Dec 26	planned	Harness / assembly house	PLACEHOLDER — RFQ line
2.8	Firmware load, end-of-line provisioning, ATECC lock	08 Dec 26	19 Dec 26	planned	Manufacturer	PLACEHOLDER — RFQ line

ID	Task	Start	End	Status	Owner	Cost basis
2.9	Per-unit test: TST-EEG-004 and TOOL-EEG-022 browser check	15 Dec 26	09 Jan 27	planned	Manufacturer	PLACEHOLDER — RFQ line
2.10	Kitting, foam, labels, flight case; ship to Brussels	05 Jan 27	16 Jan 27	planned	Manufacturer	PLACEHOLDER — RFQ line
2.11	Goods-in acceptance and correction loop (RFQ §9.4)	19 Jan 27	13 Feb 27	planned	Programme lead + manufacturer	PLACEHOLDER — per-iteration rate
2.12	Phase 2 fleet build: ten kits	01 Apr 27	30 Jun 27	planned	Manufacturer	PLACEHOLDER — RFQ line

Platform

The server platform, from frame ingest to the session report. The initial production release is the January 2027 milestone.

ID	Task	Start	End	Status	Owner	Cost basis
3.1	Server platform architecture and data-custody design	08 Sep 26	03 Oct 26	planned	Software (volunteer / contract)	In-kind, else €70–110/h
3.2	Ingest and quality control: COBS frame reception, CRC, gap handling	06 Oct 26	07 Nov 26	planned	Software	In-kind, else €70–110/h
3.3	Storage, hashing and timestamped custody chain	20 Oct 26	21 Nov 26	planned	Software	In-kind, else €70–110/h
3.4	Processing pipeline: filtering, artefact rejection, epoching	03 Nov 26	12 Dec 26	planned	Software + statistician	In-kind, else €70–110/h
3.5	Feature and model layer; classifier train/apply harness	24 Nov 26	24 Dec 26	planned	Software + statistician	In-kind, else €70–110/h
3.6	Participant-facing session module (Part 14 interface rules)	13 Oct 26	12 Dec 26	planned	Software	In-kind, else €70–110/h
3.7	Session report and Document Library deposit	01 Dec 26	24 Dec 26	planned	Software	In-kind, else €70–110/h
3.8	INITIAL PRODUCTION RELEASE of the platform	05 Jan 27	29 Jan 27	planned	Software	In-kind, else €70–110/h
3.9	Confirmatory-analysis and reporting layer	01 Feb 27	26 Mar 27	planned	Software + statistician	In-kind, else €70–110/h

Funding

Stage costs are Part 18's indicative figures. Stage 1 is the one that must be funded rather than recovered from contributions.

ID	Task	Start	End	Status	Owner	Cost basis
4.1	Detailed budget build from RFQ responses and Part 18 figures	08 Sep 26	03 Oct 26	in progress	Programme lead	—
4.2	Stage 0 funding secured (€2,150 indicative)	15 Sep 26	31 Oct 26	planned	Programme lead	€2,150 target
4.3	Stage 1 funding case and grant applications (€10,900 indicative)	01 Oct 26	29 Jan 27	planned	Programme lead	€10,900 target
4.4	Stage 2 gap funding (€7,900 of €22,900 under Model 1)	01 Mar 27	30 Jul 27	planned	Programme lead	€7,900 target
4.5	Hardship fund: open-ended participant accessibility	02 Nov 26	31 Aug 29	planned	Programme lead	Open-ended

Academic

The partnership, the ethics route, the statistics and the publications. This is the workstream that most changes the programme's cost base.

ID	Task	Start	End	Status	Owner	Cost basis
5.1	Identify and approach candidate academic / clinical partners	08 Sep 26	28 Nov 26	in progress	Programme lead	—
5.2	Partner agreement: authorship, laboratory tier, data governance	02 Nov 26	19 Dec 26	planned	Programme lead + partner	Nil under Scenario A
5.3	Ethics submission to the partner's committee	01 Dec 26	26 Feb 27	planned	Academic partner (PI)	Nil if institutional; €3,000–8,000 commercial
5.4	Statistical review of the analysis plan and pre-registration	17 Nov 26	15 Jan 27	planned	Statistician	€1,500–3,000 (inside Stage 1)
5.5	Pre-registration lodged before any Stage 1 data is collected	18 Jan 27	12 Feb 27	planned	Academic partner + statistician	—
5.6	Physician engagement: screening, on-call cover, debrief wording	01 Oct 26	31 Mar 27	planned	Hospital consultant	Nil (A) / €3,000–6,000 (B) / €9,000–22,000 (C)
5.7	Clinical neurophysiology review of montage and artefact policy	03 Nov 26	12 Dec 26	planned	Neurophysiology technician	€40–60/h loaded, €60–100/h consultancy
5.8	Conference abstract and preprint of the Stage 1 protocol	01 Mar 27	30 Apr 27	planned	Academic partner + programme lead	—
5.9	Peer-reviewed publication of Stage 1 results	01 Oct 27	28 Apr 28	planned	Academic partner (PI)	—
5.10	Stage 2 publication and open dataset release	08 Jan 29	29 Jun 29	planned	Academic partner (PI)	—

Study

The three stages of the experiment itself, each gated on the instrument, the platform and the ethics approval.

ID	Task	Start	End	Status	Owner	Cost basis
6.1	Stage 0 — proof of protocol, six to eight pilot participants	15 Feb 27	30 Apr 27	planned	Programme lead + participant support	€2,150 indicative
6.2	Stage 0 findings; protocol and instrument corrections	03 May 27	28 May 27	planned	Programme lead	—
6.3	Participant recruitment and screening for Stage 1	01 Apr 27	30 Jul 27	planned	Participant support + physician	Screening inside Stage 1
6.4	Stage 1 — the decisive study, seventy-five sessions	01 Jul 27	29 Feb 28	planned	Programme lead + participant support	€10,900 indicative
6.5	Stage 1 analysis, blinded until the cohort closes	01 Mar 28	30 Jun 28	planned	Statistician + academic partner	—
6.6	Individual participant reports issued after cohort close	03 Jul 28	31 Aug 28	planned	Participant support	—
6.7	Stage 2 — the fleet, distributed sessions at scale	01 Sep 28	29 Jun 29	planned	Programme lead + participant support	€22,900, €7,900 gap
6.8	Stage 2 analysis, reporting and dataset publication	01 May 29	31 Aug 29	planned	Statistician + academic partner	—

Operations

Running a fleet in participants' homes: logistics, refurbishment, regulatory and data protection.

ID	Task	Start	End	Status	Owner	Cost basis
7.1	Participant-support role: onboarding, scheduling, debriefs	01 Feb 27	31 Aug 29	planned	Participant support	€25–40/h if not in-kind
7.2	Kit logistics: shipping, returns, refurbishment cycle	15 Feb 27	31 Aug 29	planned	Operations / logistics	€400 Stage 0; per-session thereafter
7.3	Service and refurbishment per SVC-EEG-013, ~25 sessions per cup set	01 Jun 27	31 Aug 29	planned	Operations	Consumables per kit
7.4	Regulatory: UN3481 lithium shipping compliance, RoHS/REACH file	02 Nov 26	19 Dec 26	planned	Regulatory adviser	€150–250/h, rarely in-kind
7.5	Data-protection: DPIA, retention policy, custody documentation	01 Dec 26	26 Feb 27	planned	Programme lead + partner DPO	Nil if institutional

Roles and cost rates

Rates are from the study design v4.2, Part 18b, Table 18b.2 — Belgian and Western European figures for 2026, stated as ranges because they vary by institution, seniority, and whether the person is paid through their employer or directly.

Two columns, because the same hour has two prices. The loaded salary cost is what the contribution is worth if an institution absorbs it. The consultancy rate is what it costs if the programme has to buy it. The plan assumes the academic and clinical hours are contributed; Part 18b prices the failure of that assumption and it is not small.

Role	Responsibility	Source	If contributed	If bought
Programme lead	Direction, funding, RFQ, partner and participant relations	Founder	Nil — founder time	Nil
Electronics and firmware engineer	Carrier, firmware, test tooling, bring-up	Volunteer / in-kind	Nil — in-kind to date	€60–90/h if bought
Mechanical engineer	Helmet frame redraw, electrode assembly, enclosure	Volunteer / in-kind	Nil — in-kind to date	€60–90/h if bought
Electrical-safety reviewer	S-02, SR-01, SR-12, applied-parts review and sign-off	Engineer	€60–90/h loaded	€90–150/h consultancy
Layout engineer (RFQ-EEG-002A)	Independent review of the routing before fabrication	External	—	PLACEHOLDER — RFQ response
Software engineer(s)	Server platform, ingest, pipeline, participant module	Volunteer / contract	Nil — in-kind assumed	€70–110/h if bought
Statistician	Analysis plan, pre-registration, confirmatory analysis	Academic	€40–50/h loaded	€80–120/h; €1,500–3,000 in Stage 1
Senior academic / PI	Institutional home, ethics sponsorship, publication	Academic partner	€60–80/h loaded	€100–200/h; 10 % buyout €8,000–11,000/yr
Postdoctoral researcher	Signal-processing and statistical review	Academic partner	€40–50/h loaded	€80–120/h
Hospital consultant (neurologist / psychiatrist)	Screening, debrief wording, clinical governance	Clinical partner	€70–100/h loaded	€150–300/h; 0.1 FTE €12,000–18,000/yr
On-call medical availability	Cover during recording windows	Clinical partner	—	€25–50/h availability; active rate on calls taken
Clinical neurophysiology technician	Montage, artefact policy, quality review	Clinical partner	€40–60/h loaded	€60–100/h
Regulatory adviser	UN3481, RoHS/REACH, device-status positioning	External	—	€150–250/h — rarely in-kind
Ethics review	Study approval	Institutional committee	Nil if institutional	€3,000–8,000 flat if commercial
Participant support	Onboarding, scheduling, debriefs, hardship administration	Programme / volunteer	Nil — in-kind assumed	€25–40/h if bought
Contract manufacturer	Board, assembly, harness, print, provisioning, test, kitting	External	—	PLACEHOLDER — RFQ-EEG-001 responses
MJF print bureau	Helmet, pod and fixture printed parts	External	—	PLACEHOLDER — print quote

Note on the physician row. The study design is explicit that this is the row hardest to justify at nil: screening, on-call availability and debrief wording are real clinical work. Under honoraria it is €3,000–6,000; at consultancy rates it is €9,000–22,000 plus on-call, or a 0.1 FTE secondment at €12,000–18,000 a year.

Manufacturing process and resources

The manufacturing route the RFQ asks a bidder to price, end to end. The programme wants a finished kit — assembled, tested, packed in its flight case — and the RFQ asks any bidder who cannot deliver every step to say so, name who could, and state their own differentiator against an integrator who can.

ID	Process step	Resource	Cost
2.1	Issue RFQ-EEG-001 Rev E to shortlisted manufacturers	Programme lead	—
2.2	RFQ responses, clarifications and bid comparison	Programme lead	—
2.3	Select manufacturer; agree correction-iteration rates (RFQ §9.4)	Programme lead	PLACEHOLDER — RFQ response
2.4	Bare-board fabrication, four-layer ENIG, with lot coupon	PCB fabricator	PLACEHOLDER — RFQ line
2.5	Carrier assembly: SMT, through-hole, module fitting	EMS / assembly house	PLACEHOLDER — RFQ line
2.6	MJF printing of the mechanical set	MJF print bureau	PLACEHOLDER — RFQ line
2.7	Harness build per WH-EEG-008; helmet assembly	Harness / assembly house	PLACEHOLDER — RFQ line
2.8	Firmware load, end-of-line provisioning, ATECC lock	Manufacturer	PLACEHOLDER — RFQ line
2.9	Per-unit test: TST-EEG-004 and TOOL-EEG-022 browser check	Manufacturer	PLACEHOLDER — RFQ line
2.10	Kitting, foam, labels, flight case; ship to Brussels	Manufacturer	PLACEHOLDER — RFQ line
2.11	Goods-in acceptance and correction loop (RFQ §9.4)	Programme lead + manufacturer	PLACEHOLDER — per-iteration rate
2.12	Phase 2 fleet build: ten kits	Manufacturer	PLACEHOLDER — RFQ line

Every cost above is a placeholder. RFQ-EEG-001 Rev E section 10 carries the pricing template these will be filled from, including the correction-iteration rates of section 9.4 — because the first prototypes of a design that has never been built will need correcting, and it is cheaper to agree the rate before the order than after the fault.

Project management risks

High-level, to be developed into a managed register once the RFQ responses and the partnership position are known.

ID	Risk	Exposure	Impact	Mitigation
R1	No academic or clinical partner is found	High	Ethics review becomes commercial (€3,000–8,000), the laboratory tier is lost, and publication credibility drops. Part 18 calls this the single most valuable recruitment target.	Approach in parallel, early, and offer authorship rather than payment.
R2	RFQ responses exceed the budget envelope	High	Every manufacturing line in this plan is a placeholder. Stage 1 cannot start without kits.	Quote at 2 / 10 / 25 / 50 breaks; keep the partial-bid route of RFQ §1.1a open so specialist bidders can price what they actually do.
R3	The first prototype does not work and correction iterations multiply	High	Schedule and budget both slip. Phase 1 is two units of a design never built.	RFQ §9.4 fixes the correction loop and the per-iteration rates before the order is placed, and asks the manufacturer to hold the units on their bench until they pass.
R4	Electrical safety review returns findings that require a board respin	Medium	Adds a fabrication cycle and cost; delays Stage 0.	Review is scheduled before fabrication, not after. S-02 is already met by design and the bias topology is already corrected.
R5	The helmet frame redraw depends on a fit measurement not yet taken	Medium	HM-01 is a carried-over form study; its geometry changes after Stage 0 fit work, and the channel runs must be re-cut to match.	Parametric channel runs are already drawn to the correct section; only the head surface is outstanding.
R6	Participant recruitment is slower than planned	Medium	Stage 1 needs seventy-five sessions; a thin pipeline extends the stage.	Recruit across both cohorts in parallel from Stage 0; hardship fund widens eligibility.
R7	Platform production release slips past January 2027	Medium	Stage 0 can run on a manual pipeline; Stage 1 cannot.	Ingest, storage and the participant module are sequenced first; analysis and reporting can follow Stage 0.
R8	Volunteer engineering capacity is withdrawn or reduced	Medium	Most of the instrument and platform work is in-kind. Buying it is €60–110/h.	Keep every deliverable generated and documented so the work is transferable; Scenario C in Part 18b prices the fallback.
R9	A safety or adverse event during a session	Low likelihood, high impact	Study stops; participant harm.	On-call medical availability, screening, the charge interlock, and a device that is not a medical device and says so.
R10	Data-protection or custody failure	Low likelihood, high impact	Loss of participant trust and legal exposure.	DPIA before Stage 1; hashed and timestamped custody chain; retention policy agreed with the partner's DPO.

Two of these deserve naming as the ones that actually govern the programme. R1, the academic partner, because it converts the largest cost lines from purchases into contributions and is the difference between a credible study and a private one. And R2 with R3 together, because the instrument is a design that has never been built: the RFQ responses will tell the programme whether Stage 1 is affordable, and the correction loop will tell it whether the schedule survives contact with the first prototype.

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